

MoEF notification welcomed by solar players

In May 2011, the Ministry of Environment and Forests (MoEF) notified that no environmental clearance is required for solar PV power projects. The notification came after the Ministry of New and Renewable Energy (MNRE) took up this issue with MoEF.

The notification stated that solar PV power projects are not covered under the ambit of EIA notification, 2006, and no environmental clearance is required for such projects.

Although the environmental experts are concerned about this move, solar players have all the reasons to rejoice. In the absence of such clearance, banks or other financial institutions have always been apprehensive about lending to these developers. Hence the industry has welcomed this clarification, as it will help developers get funding as well as complete their projects in time. However, projects coming up in forest areas will have to apply for forest clearance from MoEF.

stand as the lowest emitter of greenhouse gases among all. These emission figures of the solar industry are only due to the energy consumption during the manufacture of value-chain components."

A research conducted by Brookhaven National Laboratory, New York, recommends cadmium telluride as one of the most promising PV technologies. But cadmium is one of the worst heavy metals. Still, researchers found that even when comparing direct emissions from the production of cadmium telluride cells with coal power plants, toxic emissions were 300 times lower.

Several studies have been conducted to compare greenhouse gas emissions of various power generation mechanisms, and all have found that renewable sources of power emit much lesser greenhouse gases than fossil fuels. Wind, nuclear and hydro-power have always emitted the least amount of greenhouse gases over the lifecycle of electricity generation. PV, being the next, is still four to six times better than coal- and gas-based systems.

Disposal. The amount of toxic chemicals used to make solar panels is also a cause for concern. For solar power to be truly green, the industry must reduce and eventually eliminate the use of toxic materials and develop environmentally sustainable practices. Also, manufacturers must take responsibility for the lifecycle impacts of their products by testing new materials and processes, adopting more recycling technology

and designing products for easy recycling. PV modules, for example, contain substances such as glass, aluminium and semiconductor materials that can be successfully recovered for use either in new PV modules or other products.

Despite the volume of waste being small at present, the industry has to be cautious in opting for recycling processes in order to prevent environmental damage. PV Cycle, a European voluntary PV recycling initiative, estimates that there will be 35,000 tonnes of PV waste by 2020. However, thankfully, companies like First Solar and SolarWorld have already introduced recycling schemes for thin-film and silicon modules.

Industry's response. Industry experts feel that for thin-film technologies, the biggest concern is safe disposal of hazardous materials. Manufacturers are opting for all the preventive measures to ensure safe disposal as most of them use fluorine-based recipes for chemical processes to mitigate greenhouse gas emission.

Impact on depleting water resources

There is also immense pressure on already depleting water resources due to vast installations of solar power plants. Water is needed in large volumes to wash solar panels as dust accumulation can affect the efficiency of solar panels, especially in dry areas like Gujarat and Rajasthan. These are states that have not only initiated solar power projects but also have huge installations lined up.

Industry's response. Experts believe that there is an urgent need to attend to this problem since water is a scarce resource. There is a need to lay down normative water use and management standards for solar plants.

Solutions to curb PV-linked pollution

Though solar cells are far from being a zero-emission technology, this does not mean that PV solar energy should not be promoted. Environmental protection is a global concern, so manufacturers should pay attention to reducing or curbing the pollution levels by following international standards and protocols when a system's total carbon footprint is calculated. A few solutions include:

Using the abatement system. As already pointed out, PV manufacturing processes produce hazardous solid byproducts and waste gases. These processes produce a mixture of toxic, reactive, ignitable and corrosive gases. In order to curb environmental pollution, hazardous gases need to be treated before being released into the atmosphere or water systems.

There are a number of solar cell technologies in use, some of which employ hazardous materials as basic constituents. The unutilised, non-consumed or unreacted gases (for example, in silicon-based technology, which uses gas) can be passed through an abatement system that works with the 'dry' or 'wet' method. This would help in neutralising the harmful effects of toxic gases.

Explaining the process, Biju P.C., manager, sales and marketing, Emvsee Photovoltaic Power, says, "In the dry process, the flammable gases are burnt with an external supply of gases like liquid petroleum gas (LPG) or methane to form a non-reactive ash. Here, the product is an oxide of silicon, which can be reused in the brick or glass industry. In the wet method, an external aid of other chemicals is required along with a mixture of water, which forms a non-reactive slurry of